

Quadratic Variation of the Constant-Coefficient Takagi Function Along Ternary Partitions

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A Research Report

This report arose from a summer research internship conducted under the supervision of Professor Rama Cont, Mathematical Institute, University of Oxford.

Summer 2026

Abstract

Let

$$x(t) = \sum_{m=0}^{\infty} 2^{-m/2} \operatorname{dist}(2^m t, \mathbb{Z}), \quad t \in [0, 1],$$

be the normalised constant-coefficient Takagi function associated with the Faber–Schauder system. This report investigates its quadratic sums along regular dyadic and ternary partitions.

Along dyadic partitions, the multiscale structure of the function is exactly aligned with the partition grid. A direct calculation recovers the identity

$$Q_{D_p}(x) = 1 - 2^{-p},$$

and hence quadratic variation equal to one. Along ternary partitions, this alignment is lost: interior ternary points are not dyadic, the limiting function does not coincide with any finite partial sum on the partition grid, and increments from distinct Faber–Schauder levels are no longer orthogonal.

Using the periodicity of dyadic residues modulo 3^p , governed by the multiplicative order

$$\operatorname{ord}_{3^p}(2) = 2 \cdot 3^{p-1},$$

we derive an exact finite representation of every ternary increment and of the corresponding quadratic sum $Q_{T_p}(x)$.

A numerical investigation of the partial sums x_n reveals a transition governed by the relative scale $n - p \log_2(3)$. After this rescaling, the transition curves exhibit an approximate data collapse. For the full limiting function, the computed ternary quadratic sums remain non-negligible but fluctuate over the investigated range. The present theoretical and numerical analysis does not determine whether $Q_{T_p}(x)$ converges as $p \rightarrow \infty$.

Keywords. Quadratic variation; Takagi functions; Faber–Schauder system; ternary partitions; multiplicative order.

1 Introduction

For regular functions, limits of Riemann sums are typically stable under changes in the refining sequence of partitions. For irregular functions, however, the geometry of the partitions may influence the limiting object. Quadratic variation provides a fundamental example of this phenomenon.

Given a continuous function $f: [0, 1] \rightarrow \mathbb{R}$ and a finite partition

$$\pi = \{0 = t_0 < t_1 < \cdots < t_N = 1\},$$

the associated quadratic sum is defined by

$$Q_\pi(f) = \sum_{j=0}^{N-1} (f(t_{j+1}) - f(t_j))^2.$$

Quadratic variation along a sequence of partitions is obtained, when the limit exists, by considering these sums along increasingly fine partitions. In probability, it is a fundamental ingredient of Itô calculus. Föllmer's pathwise approach showed that quadratic variation may also be studied deterministically, as a property of a path relative to a prescribed sequence of partitions [3]. The dependence of this quantity on the chosen refining sequence has subsequently been investigated in greater generality; see, for example, Cont and Das [2]. This leads to the central question of the present report: how does the limiting behaviour of quadratic sums depend on the geometry of the underlying partitions?

We investigate this question for the normalised constant-coefficient Takagi function

$$x(t) = \sum_{m=0}^{\infty} \sum_{k=0}^{2^m-1} e_{m,k}(t) = \sum_{m=0}^{\infty} 2^{-m/2} \varphi(2^m t), \quad \varphi(u) := \text{dist}(u, \mathbb{Z}).$$

where $e_{m,k}$ denotes the normalised Faber–Schauder function at level m and position k . This function belongs to a generalised Takagi class related to the classical construction introduced by Takagi in 1903 [6]; comprehensive surveys of the classical Takagi function are given in [1, 4]. The distinction between the two functions is important. The classical Takagi function uses the level weights 2^{-m} , whereas the normalisation considered here uses $2^{-m/2}$. The latter is the natural scaling for non-trivial quadratic variation and belongs to the class studied by Schied [5].

The Faber–Schauder representation decomposes x into successive dyadic scales. It is therefore naturally adapted to the regular dyadic partitions

$$D_p = \left\{ \frac{j}{2^p} : 0 \leq j \leq 2^p \right\}.$$

Along these partitions, the basis functions are affine on the relevant partition intervals and the mixed contributions from different levels cancel after summation. In the constant-coefficient case, this yields

$$Q_{D_p}(x) = 1 - 2^{-p},$$

and hence

$$\lim_{p \rightarrow \infty} Q_{D_p}(x) = 1.$$

The linear dyadic quadratic variation of this normalised generalised Takagi class is known [5, Proposition 2.6]. A direct proof is included in Section 3, both for completeness and to provide a reference case for the ternary problem.

The purpose of this report is to determine how this structure changes when the dyadic partitions are replaced by the regular ternary partitions

$$T_p = \left\{ \frac{j}{3^p} : 0 \leq j \leq 3^p \right\}.$$

This change destroys the alignment between the partition points and the dyadic breakpoints of the Faber–Schauder functions. In particular, the tail of the infinite expansion no longer vanishes at the interior points of T_p , and the cancellation argument available in the dyadic setting cannot be transferred directly.

The main theoretical contribution of this report is an exact finite representation of the increments of x along each fixed ternary partition. The key observation is that the sequence

$$m \mapsto \varphi\left(\frac{2^m j}{3^p}\right)$$

is periodic. The multiplicative order of 2 modulo 3^p is

$$L_p = \text{ord}_{3^p}(2) = 2 \cdot 3^{p-1}.$$

Since φ is even, the effective common period of the distance-function orbit is the shorter quantity

$$H_p = \frac{L_p}{2} = 3^{p-1}.$$

This arithmetic structure allows the infinitely many Faber–Schauder levels contributing at a ternary point to be grouped into finitely many geometric series. It therefore reduces the evaluation of the increments of x , and hence of $Q_{T_p}(x)$, to an exact finite computation.

This representation is then used to investigate numerically the two-parameter family

$$Q_{T_p}(x_n),$$

where x_n is the n -th partial sum of the Faber–Schauder expansion. The computations indicate that the transition towards the fixed- p limiting value occurs when the smallest dyadic scale and the ternary mesh size become comparable:

$$2^{-n} \approx 3^{-p}, \quad \text{or equivalently} \quad n \approx p \log_2(3).$$

After recentering by $n - p \log_2(3)$, the curves corresponding to different partition levels exhibit an approximate collapse onto a common transition profile. For the full function, the computed values $Q_{T_p}(x)$ remain non-negligible but fluctuate as p increases. The theoretical and numerical evidence obtained in this report does not settle whether the sequence $Q_{T_p}(x)$ converges as $p \rightarrow \infty$.

More broadly, the comparison between the dyadic and ternary settings illustrates that the quadratic variation of an irregular deterministic function need not be determined by the path alone. It may depend essentially on the interaction between the multiscale geometry of the function and the arithmetic structure of the partitions along which it is sampled.

The report is organised as follows. Section 2 introduces the normalised constant-coefficient Takagi function and its Faber–Schauder representation. Section 3 treats the dyadic case. Section 4 identifies the obstruction created by ternary partitions. Section 5 derives the exact finite representation of the ternary increments. Section 6 presents the numerical investigation, and Section 7 discusses the results and the remaining open questions.

2 The Constant-Coefficient Takagi Function

This section introduces the function studied throughout the report. We first recall the normalised Faber–Schauder system and the corresponding class of generalised Takagi functions. We then specialise to the constant-coefficient choice

$$\theta_{m,k} = 1.$$

2.1 The Faber–Schauder system

Let

$$e_{0,0}(t) = (\min\{t, 1 - t\})^+, \quad t \in \mathbb{R},$$

where $a^+ = \max\{a, 0\}$. Thus, $e_{0,0}$ is the tent function supported on $[0, 1]$, with maximum value $1/2$ attained at $t = 1/2$.

For every integer $m \geq 0$ and $k \in \mathbb{Z}$, define

$$e_{m,k}(t) = 2^{-m/2} e_{0,0}(2^m t - k).$$

The function $e_{m,k}$ is supported on the dyadic interval

$$\left[\frac{k}{2^m}, \frac{k+1}{2^m} \right],$$

and reaches its maximum value

$$2^{-m/2-1}$$

at the midpoint $(k+1/2)2^{-m}$. On each of the two halves of its support, its slope is respectively $2^{m/2}$ and $-2^{m/2}$.

When the functions are restricted to $[0, 1]$, the relevant indices at level m are

$$k = 0, \dots, 2^m - 1.$$

The collection

$$\{e_{m,k} : m \geq 0, 0 \leq k \leq 2^m - 1\}$$

is the normalised Faber–Schauder system on $[0, 1]$.

The geometry of a general Faber–Schauder function is illustrated in Figure 1.

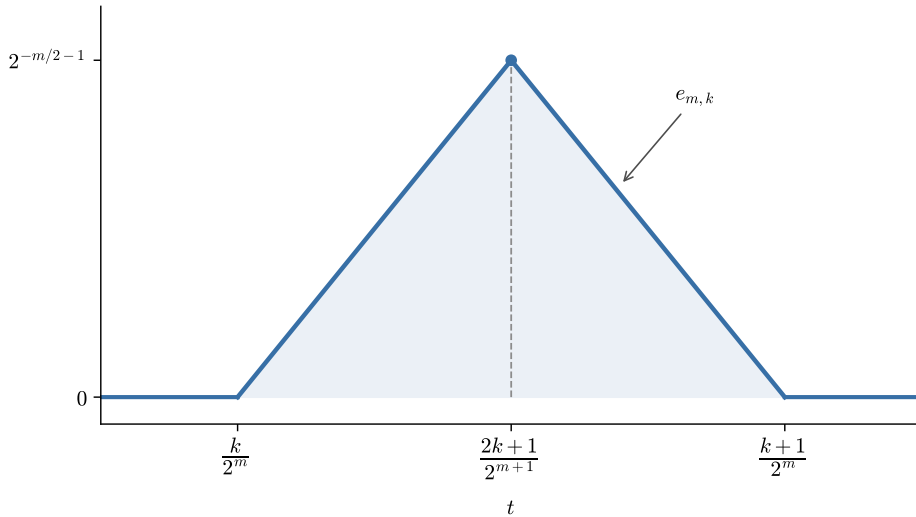


Figure 1: Geometry of the normalised Faber–Schauder function $e_{m,k}$. Its support is the dyadic interval $[k2^{-m}, (k+1)2^{-m}]$, and its maximum value is $2^{-m/2-1}$.

On the interval $[0, 1]$, the mother tent function satisfies

$$e_{0,0}(t) = \varphi(t),$$

where $\varphi(u) = \text{dist}(u, \mathbb{Z})$. The distinction between these two functions outside $[0, 1]$ should be noted: $e_{0,0}$ is compactly supported, whereas φ is 1-periodic on $\mathbb{R}$.

2.2 A normalised Takagi class

Let

$$\theta_{m,k} \in \{-1, 1\}, \quad m \geq 0, \quad 0 \leq k \leq 2^m - 1.$$

For every integer $n \geq 1$, define the truncated Takagi function

$$x_n^\theta(t) = \sum_{m=0}^{n-1} \sum_{k=0}^{2^m-1} \theta_{m,k} e_{m,k}(t), \quad 0 \leq t \leq 1.$$

We refer to x^θ as the normalised Takagi function associated with the coefficient family $\theta = (\theta_{m,k})$.

For each fixed $t \in [0, 1]$ and each level m , at most one function $e_{m,k}(t)$ is non-zero. Consequently,

$$\left\| \sum_{k=0}^{2^m-1} \theta_{m,k} e_{m,k} \right\|_\infty \leq 2^{-m/2-1}.$$

Since

$$\sum_{m=0}^{\infty} 2^{-m/2-1} < \infty,$$

the Weierstrass M -test shows that the sequence $(x_n^\theta)_{n \geq 1}$ converges uniformly on $[0, 1]$. Its limit is therefore continuous and is given by

$$x^\theta(t) = \sum_{m=0}^{\infty} \sum_{k=0}^{2^m-1} \theta_{m,k} e_{m,k}(t).$$

In this report, we study the constant-coefficient choice

$$\theta_{m,k} = 1$$

for every m and k . We consequently write

$$x_n(t) = \sum_{m=0}^{n-1} \sum_{k=0}^{2^m-1} e_{m,k}(t)$$

and

$$x(t) = \sum_{m=0}^{\infty} \sum_{k=0}^{2^m-1} e_{m,k}(t) = \lim_{n \rightarrow \infty} x_n(t),$$

where the convergence is uniform on $[0, 1]$. The function x will be referred to as the constant-coefficient Takagi function.

Figure 2 illustrates the uniform construction of x . The functions x_n are non-negative and increase pointwise with n , since every Faber–Schauder coefficient is equal to 1. At the same time, the amplitude of the oscillations introduced at level m decreases geometrically.

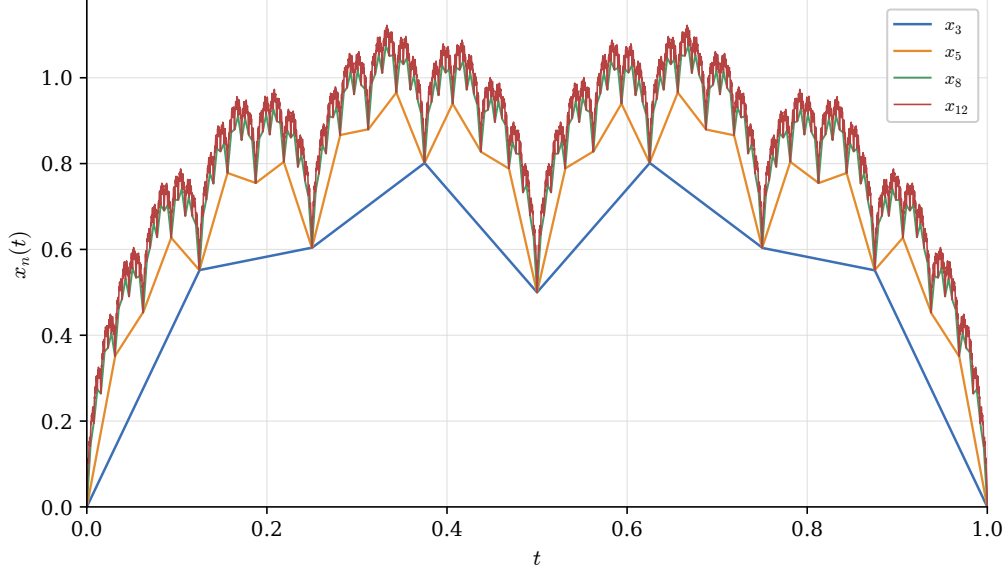


Figure 2: Successive truncations x_3 , x_5 , x_8 , and x_{12} of the constant-coefficient Takagi function. Each additional level introduces finer non-negative dyadic oscillations of decreasing amplitude.

2.3 Dyadic multiscale structure

The Faber–Schauder expansion gives x an explicitly dyadic multiscale structure. At level m , the supports of the basis functions have length 2^{-m} , while their heights are equal to $2^{-m/2-1}$. Increasing m therefore produces oscillations that are more localised but have increasingly steep slopes.

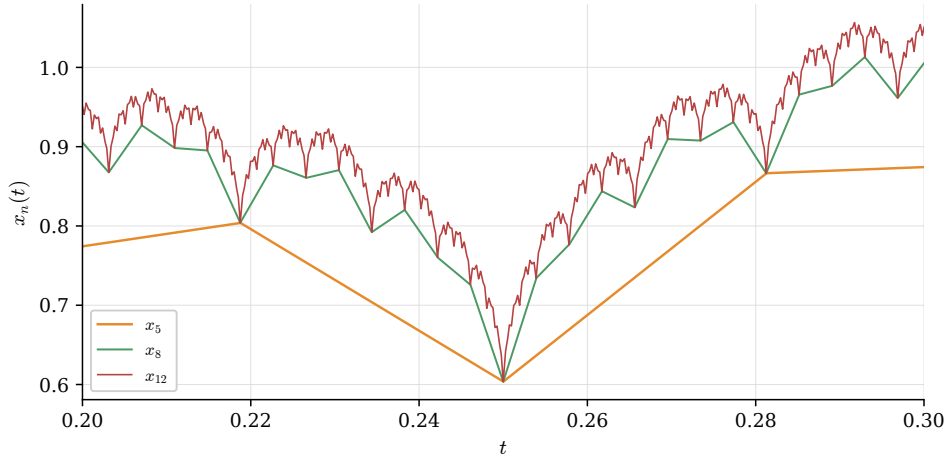


Figure 3: Local view of the truncations x_5 , x_8 , and x_{12} on a subinterval of $[0, 1]$. The successive refinements reveal the hierarchical dyadic organisation of the Faber–Schauder expansion.

The local view in Figure 3 makes this hierarchy particularly visible. Passing from x_n to x_{n+1} adds a new family of tents supported on dyadic intervals of length 2^{-n} .

This dyadic organisation is naturally compatible with dyadic partitions. Its interaction with ternary partitions is more subtle: neither family of regular grids refines the other, and the dyadic and ternary nodes intersect only at 0 and 1. This incompatibility will play a central role in the following sections.

3 Quadratic Variation Along Dyadic Partitions

Before turning to ternary partitions, we examine the quadratic variation of the normalised constant-coefficient Takagi function along dyadic partitions. In this setting, the partition grid is exactly aligned with the Faber–Schauder expansion, which leads to an explicit calculation.

The existence of linear quadratic variation for the wider normalised Takagi class is known; see Schied [5, Proposition 2.6]. We include the direct constant-coefficient calculation because it provides both the exact finite-level identity and the reference case for the ternary problem.

3.1 Quadratic variation and dyadic partitions

Let

$$\pi = \{0 = t_0 < t_1 < \cdots < t_N = 1\}$$

be a finite partition of $[0, 1]$. For a continuous function $f : [0, 1] \rightarrow \mathbb{R}$, we define its quadratic sum along π by

$$Q_\pi(f) = \sum_{j=0}^{N-1} (f(t_{j+1}) - f(t_j))^2.$$

For every integer $p \geq 0$, let

$$D_p = \left\{ \frac{j}{2^p} : 0 \leq j \leq 2^p \right\}$$

denote the dyadic partition of level p . Its mesh size is

$$|D_p| = 2^{-p},$$

which tends to zero as $p \rightarrow \infty$.

For convenience, define the increment

$$\Delta_{p,j}f = f\left(\frac{j+1}{2^p}\right) - f\left(\frac{j}{2^p}\right), \quad 0 \leq j \leq 2^p - 1.$$

The quadratic sum of f along D_p can then be written as

$$Q_{D_p}(f) = \sum_{j=0}^{2^p-1} (\Delta_{p,j}f)^2.$$

If the limit exists, the quadratic variation of f along the sequence of dyadic partitions is defined by

$$Q_D(f) = \lim_{p \rightarrow \infty} Q_{D_p}(f).$$

3.2 Exact computation

Recall that

$$x_n(t) = \sum_{m=0}^{n-1} \sum_{k=0}^{2^m-1} e_{m,k}(t)$$

is the n -th partial sum of the constant-coefficient Takagi function.

For each $m \geq 0$, set

$$E_m(t) = \sum_{k=0}^{2^m-1} e_{m,k}(t),$$

so that

$$x_n = \sum_{m=0}^{n-1} E_m.$$

The functions E_m describe the successive dyadic levels of the Faber–Schauder expansion. When $m < p$, every breakpoint of E_m belongs to D_p . Hence E_m is affine on each interval of D_p , with slope equal to either $2^{m/2}$ or $-2^{m/2}$.

Theorem 3.1 (Dyadic quadratic variation). *Let $p \geq 1$ and $1 \leq n \leq p$. Then*

$$Q_{D_p}(x_n) = 2^{-p}(2^n - 1) = 2^{n-p}(1 - 2^{-n}).$$

Moreover,

$$Q_{D_p}(x) = Q_{D_p}(x_p) = 1 - 2^{-p}.$$

Consequently, the normalised constant-coefficient Takagi function has quadratic variation equal to 1 along the dyadic partitions:

$$Q_D(x) = \lim_{p \rightarrow \infty} Q_{D_p}(x) = 1.$$

Proof. Fix integers $1 \leq n \leq p$. For every $m \in \{0, \dots, n-1\}$ and every $j \in \{0, \dots, 2^p-1\}$, the function E_m is affine on

$$I_{p,j} = \left[\frac{j}{2^p}, \frac{j+1}{2^p} \right].$$

Its slope on this interval is either $2^{m/2}$ or $-2^{m/2}$. We may therefore write

$$\Delta_{p,j} E_m = 2^{-p} 2^{m/2} \varepsilon_{m,j},$$

where

$$\varepsilon_{m,j} \in \{-1, 1\}.$$

Since $x_n = \sum_{m=0}^{n-1} E_m$, it follows that

$$\Delta_{p,j} x_n = 2^{-p} \sum_{m=0}^{n-1} 2^{m/2} \varepsilon_{m,j}.$$

Consequently,

$$Q_{D_p}(x_n) = 2^{-2p} \sum_{j=0}^{2^p-1} \left(\sum_{m=0}^{n-1} 2^{m/2} \varepsilon_{m,j} \right)^2.$$

Expanding the square gives

$$\begin{aligned} Q_{D_p}(x_n) &= 2^{-2p} \sum_{m=0}^{n-1} 2^m \sum_{j=0}^{2^p-1} \varepsilon_{m,j}^2 \\ &\quad + 2^{1-2p} \sum_{0 \leq m < \ell < n} 2^{(m+\ell)/2} \sum_{j=0}^{2^p-1} \varepsilon_{m,j} \varepsilon_{\ell,j}. \end{aligned}$$

We first consider the mixed terms. Fix $0 \leq m < \ell < n$. The sign $\varepsilon_{m,j}$ is constant on every block of 2^{p-m-1} consecutive intervals of D_p .

On the other hand, the sign pattern $\varepsilon_{\ell,j}$ has period $2^{p-\ell}$, and its sum over each period is zero. Since

$$2^{p-m-1} = 2^{\ell-m-1} 2^{p-\ell},$$

each block on which $\varepsilon_{m,j}$ is constant contains an integer number of complete periods of the finer sign pattern. Therefore,

$$\sum_{j=0}^{2^p-1} \varepsilon_{m,j} \varepsilon_{\ell,j} = 0.$$

Thus, all mixed terms vanish.

Since $\varepsilon_{m,j}^2 = 1$, the remaining diagonal terms give

$$\begin{aligned} Q_{D_p}(x_n) &= 2^{-2p} \sum_{m=0}^{n-1} 2^m \sum_{j=0}^{2^p-1} 1 \\ &= 2^{-2p} \sum_{m=0}^{n-1} 2^m 2^p \\ &= 2^{-p} \sum_{m=0}^{n-1} 2^m \\ &= 2^{-p} (2^n - 1) \\ &= 2^{n-p} (1 - 2^{-n}). \end{aligned}$$

It remains to pass from the partial sums to the limiting function. Let

$$t = \frac{j}{2^p} \in D_p.$$

For every $m \geq p$ and every $k \in \{0, \dots, 2^m - 1\}$, we have

$$2^m t - k = j 2^{m-p} - k \in \mathbb{Z}.$$

Since the mother tent function $e_{0,0}$ vanishes at every integer,

$$e_{m,k}(t) = 2^{-m/2} e_{0,0}(2^m t - k) = 0.$$

Therefore, all levels $m \geq p$ vanish at the points of D_p , and

$$x(t) = x_p(t), \quad t \in D_p.$$

The functions x and x_p consequently have identical increments along D_p . Hence

$$Q_{D_p}(x) = Q_{D_p}(x_p).$$

Applying the first part of the proof with $n = p$, we obtain

$$Q_{D_p}(x) = 2^{-p} (2^p - 1) = 1 - 2^{-p}.$$

Finally,

$$Q_D(x) = \lim_{p \rightarrow \infty} Q_{D_p}(x) = \lim_{p \rightarrow \infty} (1 - 2^{-p}) = 1.$$

□

The dyadic computation rests on three compatible features. First, every Faber–Schauder level is affine on the intervals of a sufficiently fine dyadic partition. Second, the increment patterns associated with distinct levels are orthogonal when summed over the partition intervals. Third, the values of the limiting function on D_p are determined exactly by the finite partial sum x_p .

The ternary partitions possess none of these three properties in the same form. The next section identifies the resulting obstructions before a different method is introduced in Section 5.

4 Why Ternary Partitions Are Different

We now replace the dyadic partitions by regular ternary partitions while keeping the normalised constant-coefficient Takagi function unchanged. This apparently minor modification destroys the exact compatibility between the partition grid and the dyadic structure of the Faber–Schauder expansion.

The purpose of this section is to identify precisely which parts of the dyadic argument fail. Two principal obstructions arise: the limiting function no longer coincides with a finite partial sum at the interior partition points, and the increment patterns associated with distinct dyadic levels are no longer orthogonal on the ternary grid.

4.1 Regular ternary partitions

For every integer $p \geq 0$, define the regular ternary partition of $[0, 1]$ by

$$T_p = \left\{ \frac{j}{3^p} : 0 \leq j \leq 3^p \right\}.$$

Its mesh size is

$$|T_p| = 3^{-p},$$

which tends to zero as $p \rightarrow \infty$.

Using the notation introduced in the previous section, the quadratic sum of a continuous function f along T_p is

$$Q_{T_p}(f) = \sum_{j=0}^{3^p-1} \left(f\left(\frac{j+1}{3^p}\right) - f\left(\frac{j}{3^p}\right) \right)^2.$$

If the limit exists, we write

$$Q_T(f) = \lim_{p \rightarrow \infty} Q_{T_p}(f).$$

Our aim is to investigate the sequence

$$(Q_{T_p}(x))_{p \geq 0}$$

for the constant-coefficient Takagi function x . At this stage, the existence of a limit is not assumed.

4.2 Loss of alignment with the Faber–Schauder system

The dyadic computation relied on the fact that the breakpoints of every Faber–Schauder function are dyadic points. More precisely, the breakpoints of $e_{m,k}$ are

$$\frac{k}{2^m}, \quad \frac{2k+1}{2^{m+1}}, \quad \frac{k+1}{2^m}.$$

Consequently, when $m < p$, the function $e_{m,k}$ is affine on every interval of the dyadic partition D_p .

This property does not extend uniformly to ternary partitions. The endpoints of a ternary interval are of the form

$$\frac{j}{3^p} \quad \text{and} \quad \frac{j+1}{3^p},$$

and, apart from 0 and 1, these points do not coincide with dyadic breakpoints. Indeed, the dyadic and ternary grids intersect only at their common endpoints.

Depending on the relative values of m and p , a ternary interval may contain zero, one, or several breakpoints of $e_{m,k}$. Consequently, $e_{m,k}$ need not be affine on

$$\left[\frac{j}{3^p}, \frac{j+1}{3^p} \right],$$

and its increment cannot in general be obtained by multiplying a single slope by the interval length 3^{-p} . The contribution of a Faber–Schauder level therefore depends on the precise position of its breakpoints relative to the ternary grid.

4.3 Failure of finite-level coincidence

A crucial step in the dyadic proof was the identity

$$x(t) = x_p(t), \quad t \in D_p.$$

There is no analogous identity on a ternary partition.

Proposition 4.1. *Let $p \geq 1$ and let*

$$t = \frac{j}{3^p}, \quad 1 \leq j \leq 3^p - 1,$$

be an interior point of T_p . Then t is not a dyadic rational. Moreover, for every $n \geq 1$,

$$x(t) > x_n(t).$$

Proof. Suppose that t were dyadic. Then there would exist integers a and $r \geq 0$ such that

$$\frac{j}{3^p} = \frac{a}{2^r}.$$

It would follow that

$$2^r j = 3^p a.$$

Since 2^r and 3^p are relatively prime, this equality implies that 3^p divides j . This is impossible because

$$1 \leq j \leq 3^p - 1.$$

Therefore, t is not dyadic.

Recall that the contribution of level m is

$$E_m(t) = \sum_{k=0}^{2^m-1} e_{m,k}(t).$$

For $t \in [0, 1]$, this can equivalently be written as

$$E_m(t) = 2^{-m/2} \varphi(2^m t).$$

Since t is not dyadic, $2^m t \notin \mathbb{Z}$ for every $m \geq 0$. Hence

$$E_m(t) > 0$$

for every $m \geq 0$.

Because all the coefficients of the Takagi function considered here are equal to 1, we obtain

$$x(t) - x_n(t) = \sum_{m=n}^{\infty} E_m(t) > 0.$$

Thus $x(t) > x_n(t)$ for every finite n . □

Consequently, no finite truncation x_n agrees with x at the interior points of T_p . In particular, $Q_{T_p}(x)$ cannot be reduced directly to the quadratic sum of a finite partial sum, as it can in the dyadic case.

4.4 Loss of orthogonality between levels

For a continuous function f , define its increment over the j -th ternary interval by

$$\Delta_{p,j}^T f = f\left(\frac{j+1}{3^p}\right) - f\left(\frac{j}{3^p}\right).$$

Since

$$x_n = \sum_{m=0}^{n-1} E_m,$$

we have

$$\Delta_{p,j}^T x_n = \sum_{m=0}^{n-1} \Delta_{p,j}^T E_m.$$

It follows that

$$Q_{T_p}(x_n) = \sum_{j=0}^{3^p-1} \left(\sum_{m=0}^{n-1} \Delta_{p,j}^T E_m \right)^2.$$

Expanding the square yields

$$\begin{aligned} Q_{T_p}(x_n) &= \sum_{m=0}^{n-1} \sum_{j=0}^{3^p-1} \left(\Delta_{p,j}^T E_m \right)^2 \\ &\quad + 2 \sum_{0 \leq m < \ell < n} \sum_{j=0}^{3^p-1} \Delta_{p,j}^T E_m \Delta_{p,j}^T E_\ell. \end{aligned}$$

The first term contains the separate contribution of each dyadic level. The second term contains the interactions between distinct levels.

Unlike in the dyadic setting, these mixed terms do not vanish systematically. This can already be seen on the first ternary partition $T_1 = \{0, 1/3, 2/3, 1\}$. Indeed, on the first ternary partition

$$T_1 = \left\{ 0, \frac{1}{3}, \frac{2}{3}, 1 \right\},$$

we have

$$(\Delta_{1,j}^T E_0)_{j=0}^2 = \left(\frac{1}{3}, 0, -\frac{1}{3}\right),$$

whereas

$$(\Delta_{1,j}^T E_1)_{j=0}^2 = \frac{1}{\sqrt{2}} \left(\frac{1}{3}, 0, -\frac{1}{3}\right).$$

Their discrete inner product is therefore

$$\sum_{j=0}^2 \Delta_{1,j}^T E_0 \Delta_{1,j}^T E_1 = \frac{2}{9\sqrt{2}} \neq 0.$$

Thus, even the first two dyadic levels fail to be orthogonal when their increments are sampled on a ternary grid. The cancellation mechanism used in the dyadic proof therefore cannot be transferred to ternary partitions.

4.5 Consequences for the ternary problem

The dyadic proof cannot be reproduced by simply replacing 2^p with 3^p . Two distinct difficulties must be addressed:

1. the limiting function x does not coincide with any finite partial sum at the interior points of T_p ;
2. the mixed interactions between distinct Faber–Schauder levels do not vanish systematically.

A different method is therefore required. In the next section, we exploit the arithmetic structure of the ternary points to derive an exact finite representation of the increments of x along each fixed partition T_p .

5 An Exact Finite Representation of the Ternary Increments

The previous section showed that the dyadic reduction

$$x(t) = x_p(t), \quad t \in D_p,$$

has no analogue on the ternary grid. At every interior point of T_p , infinitely many Faber–Schauder levels contribute to the value of the limiting function.

Nevertheless, for each fixed ternary partition, these infinitely many contributions possess a periodic arithmetic structure. This periodicity allows the infinite tail to be summed exactly and leads to a finite formula for every ternary increment of x .

5.1 Distance-function representation

Recall the 1-periodic distance function

$$\varphi(u) = \text{dist}(u, \mathbb{Z}) = \min_{q \in \mathbb{Z}} |u - q|, \quad u \in \mathbb{R}.$$

It satisfies

$$0 \leq \varphi(u) \leq \frac{1}{2}, \quad \varphi(-u) = \varphi(u).$$

Recall that the contribution of the m -th Faber–Schauder level is

$$E_m(t) = \sum_{k=0}^{2^m-1} e_{m,k}(t).$$

For each fixed $t \in [0, 1]$, at most one term in this sum is non-zero. Using the definition of $e_{m,k}$, we obtain

$$E_m(t) = 2^{-m/2} \varphi(2^m t).$$

Consequently, the constant-coefficient Takagi function admits the representation

$$x(t) = \sum_{m=0}^{\infty} 2^{-m/2} \varphi(2^m t),$$

while its n -th partial sum is

$$x_n(t) = \sum_{m=0}^{n-1} 2^{-m/2} \varphi(2^m t).$$

This representation separates the contribution of the m -th level into the amplitude factor $2^{-m/2}$ and the geometric term $\varphi(2^m t)$.

5.2 Arithmetic periodicity on a ternary grid

Fix $p \geq 1$ and define

$$L_p = 2 \cdot 3^{p-1}.$$

We first identify L_p as the multiplicative order of 2 modulo 3^p .

Lemma 5.1 (Multiplicative order). *For every integer $p \geq 1$,*

$$\text{ord}_{3^p}(2) = 2 \cdot 3^{p-1}.$$

Equivalently,

$$2^{L_p} \equiv 1 \pmod{3^p},$$

and L_p is the smallest positive integer satisfying this congruence.

Proof. We first prove that, for every integer $q \geq 0$,

$$v_3(2^{2 \cdot 3^q} - 1) = q + 1,$$

where v_3 denotes the 3-adic valuation.

For $q = 0$,

$$2^2 - 1 = 3,$$

so the identity holds. Suppose that

$$v_3(2^{2 \cdot 3^q} - 1) = q + 1,$$

and set

$$A = 2^{2 \cdot 3^q}.$$

Then

$$2^{2 \cdot 3^{q+1}} - 1 = A^3 - 1 = (A - 1)(A^2 + A + 1).$$

Since $A \equiv 1 \pmod{3}$, we may write $A = 1 + 3b$ for some integer b . Consequently,

$$A^2 + A + 1 = 3(1 + 3b + 3b^2),$$

and the factor $1 + 3b + 3b^2$ is not divisible by 3. Hence

$$v_3(A^2 + A + 1) = 1.$$

It follows that

$$v_3(2^{2 \cdot 3^{q+1}} - 1) = v_3(A - 1) + 1 = q + 2.$$

The valuation identity follows by induction. Taking $q = p - 1$ gives

$$v_3(2^{2 \cdot 3^{p-1}} - 1) = p,$$

and therefore

$$2^{2 \cdot 3^{p-1}} \equiv 1 \pmod{3^p}.$$

It remains to prove minimality. By Lagrange's theorem, the order of 2 modulo 3^p divides

$$|(\mathbb{Z}/3^p\mathbb{Z})^\times| = 2 \cdot 3^{p-1}.$$

Since the order of 2 modulo 3 is 2, its order modulo 3^p must be of the form

$$2 \cdot 3^q$$

for some $0 \leq q \leq p - 1$. The valuation identity shows that

$$3^p \mid 2^{2 \cdot 3^q} - 1$$

only if $q + 1 \geq p$. Thus $q = p - 1$, and hence

$$\text{ord}_{3^p}(2) = 2 \cdot 3^{p-1}.$$

□

The multiplicative order gives a period for the residues of powers of 2. The symmetry of the distance function allows this period to be divided by two.

Corollary 5.2 (Period of the distance-function orbit). *Let*

$$H_p = 3^{p-1} = \frac{L_p}{2}.$$

Then

$$2^{H_p} \equiv -1 \pmod{3^p}.$$

Consequently, for every $j \in \{0, \dots, 3^p\}$ and every $m \geq 0$,

$$\varphi\left(\frac{2^{m+H_p}j}{3^p}\right) = \varphi\left(\frac{2^m j}{3^p}\right).$$

Thus the sequence

$$m \mapsto \varphi\left(\frac{2^m j}{3^p}\right)$$

is periodic with period dividing H_p .

Proof. By Lemma 5.1,

$$2^{2H_p} \equiv 1 \pmod{3^p},$$

whereas

$$2^{H_p} \not\equiv 1 \pmod{3^p},$$

because the order of 2 is $2H_p$. Set $A = 2^{H_p}$. Then

$$3^p \mid (A - 1)(A + 1).$$

Since

$$\gcd(A - 1, A + 1) \mid 2,$$

the two factors have no common factor of 3. Therefore 3^p divides one of them. It cannot divide $A - 1$, so

$$A \equiv -1 \pmod{3^p}.$$

Hence there exists an integer b_p such that

$$2^{H_p} = -1 + b_p 3^p.$$

For every $m \geq 0$,

$$\frac{2^{m+H_p} j}{3^p} = -\frac{2^m j}{3^p} + b_p 2^m j.$$

The second term is an integer. Since φ is 1-periodic and even,

$$\varphi\left(\frac{2^{m+H_p} j}{3^p}\right) = \varphi\left(-\frac{2^m j}{3^p}\right) = \varphi\left(\frac{2^m j}{3^p}\right).$$

□

5.3 Exact representation of the increments

For $j = 0, \dots, 3^p - 1$, recall the ternary increment

$$\Delta_{p,j}^T x = x\left(\frac{j+1}{3^p}\right) - x\left(\frac{j}{3^p}\right).$$

For every $m \geq 0$, define

$$d_{p,j}(m) = \varphi\left(\frac{2^m(j+1)}{3^p}\right) - \varphi\left(\frac{2^m j}{3^p}\right).$$

The quantity $d_{p,j}(m)$ records the geometric contribution of the m -th dyadic level to the increment over

$$\left[\frac{j}{3^p}, \frac{j+1}{3^p}\right].$$

By Corollary 5.2,

$$d_{p,j}(m + H_p) = d_{p,j}(m).$$

We can now state the main exact result.

Theorem 5.3 (Exact finite representation of ternary increments). *Let $p \geq 1$ and set*

$$H_p = 3^{p-1}.$$

Then, for every $j = 0, \dots, 3^p - 1$,

$$\Delta_{p,j}^T x = \frac{1}{1 - 2^{-H_p/2}} \sum_{r=0}^{H_p-1} 2^{-r/2} d_{p,j}(r),$$

where

$$d_{p,j}(r) = \varphi\left(\frac{2^r(j+1)}{3^p}\right) - \varphi\left(\frac{2^r j}{3^p}\right).$$

Proof. Using the distance-function representation of x , we obtain

$$\begin{aligned} \Delta_{p,j}^T x &= \sum_{m=0}^{\infty} 2^{-m/2} \left[\varphi\left(\frac{2^m(j+1)}{3^p}\right) - \varphi\left(\frac{2^m j}{3^p}\right) \right] \\ &= \sum_{m=0}^{\infty} 2^{-m/2} d_{p,j}(m). \end{aligned}$$

Since

$$|d_{p,j}(m)| \leq \frac{1}{2},$$

this series is absolutely convergent and may be regrouped without changing its value. Every non-negative integer m can be written uniquely as

$$m = qH_p + r, \quad q \geq 0, \quad 0 \leq r \leq H_p - 1.$$

Consequently,

$$\Delta_{p,j}^T x = \sum_{q=0}^{\infty} \sum_{r=0}^{H_p-1} 2^{-(qH_p+r)/2} d_{p,j}(qH_p + r).$$

By periodicity,

$$d_{p,j}(qH_p + r) = d_{p,j}(r),$$

and therefore

$$\begin{aligned} \Delta_{p,j}^T x &= \left(\sum_{q=0}^{\infty} 2^{-qH_p/2} \right) \left(\sum_{r=0}^{H_p-1} 2^{-r/2} d_{p,j}(r) \right) \\ &= \frac{1}{1 - 2^{-H_p/2}} \sum_{r=0}^{H_p-1} 2^{-r/2} d_{p,j}(r). \end{aligned}$$

□

The formula gives the exact increment of the limiting function x . It does not involve a truncation x_n and therefore contains no truncation error.

5.4 Exact representation of the quadratic sum

Substituting the preceding increment formula into the definition of $Q_{T_p}(x)$ gives the following finite expression.

Corollary 5.4 (Exact finite representation of $Q_{T_p}(x)$). *For every $p \geq 1$,*

$$Q_{T_p}(x) = \frac{1}{(1 - 2^{-H_p/2})^2} \sum_{j=0}^{3^p-1} \left(\sum_{r=0}^{H_p-1} 2^{-r/2} d_{p,j}(r) \right)^2,$$

where

$$H_p = 3^{p-1}$$

and

$$d_{p,j}(r) = \varphi\left(\frac{2^r(j+1)}{3^p}\right) - \varphi\left(\frac{2^r j}{3^p}\right).$$

Proof. By Theorem 5.3,

$$\Delta_{p,j}^T x = \frac{1}{1 - 2^{-H_p/2}} \sum_{r=0}^{H_p-1} 2^{-r/2} d_{p,j}(r).$$

Squaring this identity and summing over $j = 0, \dots, 3^p - 1$ gives the stated formula. $\square$

For each fixed p , the original problem involves the infinite series defining x . Corollary 5.4 reduces it to a finite calculation involving a $3^p \times H_p$ array, with

$$3^p H_p = 3^{2p-1}.$$

The representation is exact and contains no truncation error. Compared with the formula based directly on the full multiplicative order L_p , the symmetry of φ reduces the number of terms by a factor of two. This representation provides the theoretical basis for the numerical investigation in the next section.

6 Numerical Investigation

The exact representation derived in Section 5 makes it possible to evaluate the quadratic sums of the full limiting function along fixed ternary partitions. Before considering x , we first investigate the two-parameter family

$$Q_{T_p}(x_n),$$

where n is the truncation level of the Takagi function and p is the refinement level of the ternary partition.

The computations are performed using integer modular arithmetic. For

$$r_{m,j} \equiv 2^m j \pmod{3^p}, \quad 0 \leq r_{m,j} < 3^p,$$

we have

$$\varphi\left(\frac{2^m j}{3^p}\right) = \frac{1}{3^p} \min\{r_{m,j}, 3^p - r_{m,j}\}.$$

The residues are generated recursively through

$$r_{m+1,j} \equiv 2r_{m,j} \pmod{3^p}.$$

This avoids the loss of fractional information that would occur if $2^m j / 3^p$ were evaluated directly for large m .

6.1 Behaviour for fixed ternary partitions

We begin by fixing p and studying the evolution of $Q_{T_p}(x_n)$ as the truncation level n increases.

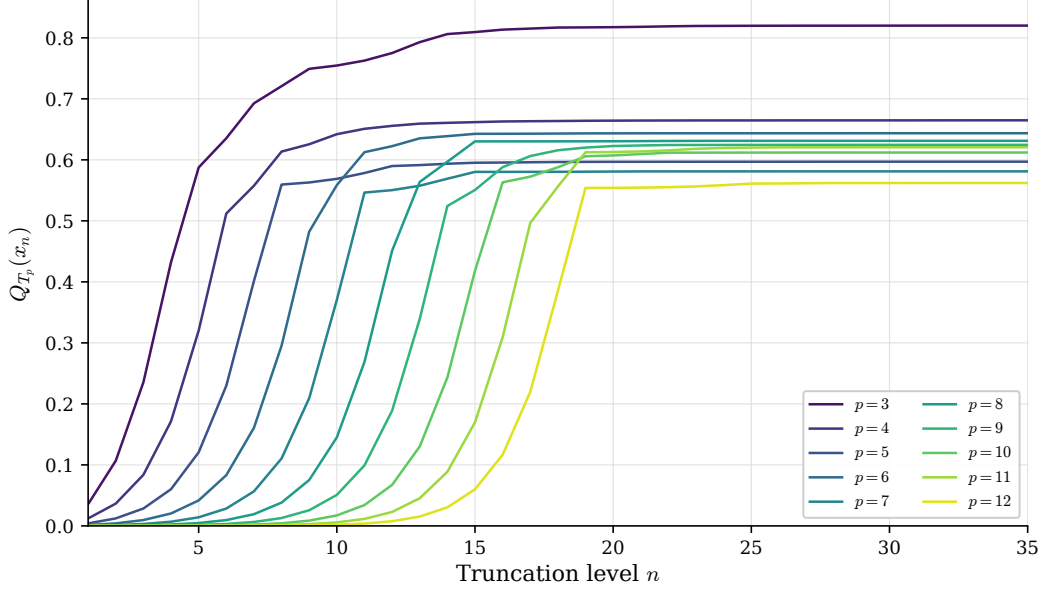


Figure 4: Quadratic sums $Q_{T_p}(x_n)$ as functions of the truncation level n , for ternary partition levels $p = 3, \dots, 12$. The transition towards the fixed- p limiting value is shifted towards larger values of n as the partition is refined.

Figure 4 reveals two distinct regimes for $p \geq 3$. When n is small relative to p , the quadratic sum is close to zero. It then increases rapidly over a relatively narrow range of truncation levels before approaching its fixed- p limiting value $Q_{T_p}(x)$.

The two coarsest partitions, $p = 1$ and $p = 2$, are omitted from this figure because their substantially larger limiting values would obscure the transition visible for the finer partitions. Their values are not discarded from the analysis and are included in Figure 7.

Although the plateau values depend on p , the most prominent effect of increasing the partition level is the displacement of the transition towards larger values of n . This indicates that the relevant parameter is not the truncation level alone, but its position relative to the resolution of the ternary partition.

6.2 Critical scaling

To study the dependence on both parameters simultaneously, we display the values of $Q_{T_p}(x_n)$ over the (n, p) -plane.

The heatmap in Figure 5 displays a clear boundary between a low-value region and a region in which the quadratic sums are close to their fixed- p limits. This boundary is approximately linear in the (n, p) -plane and closely follows the relation

$$n = p \log_2(3).$$

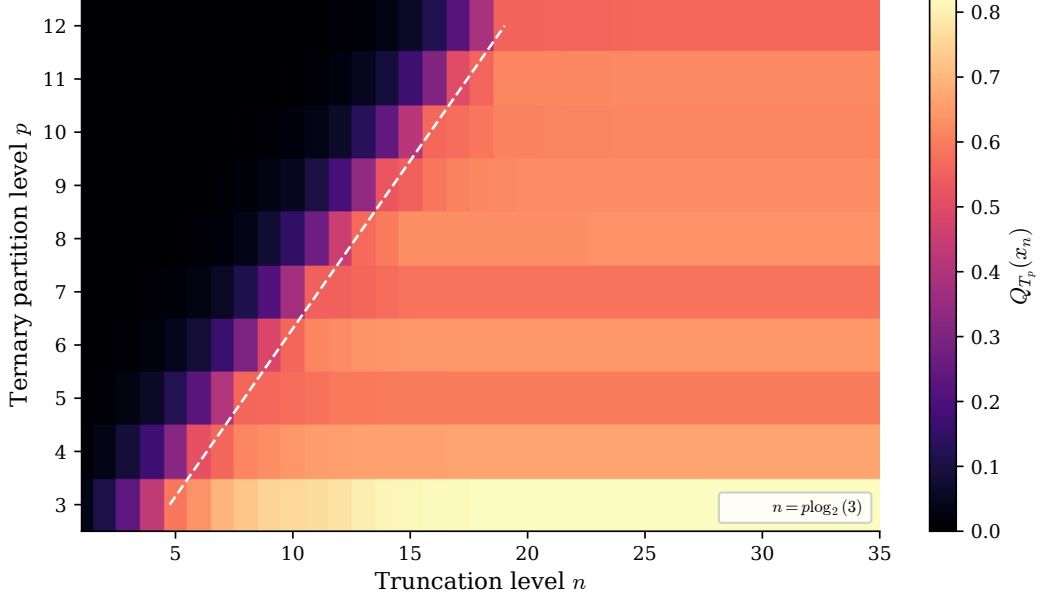


Figure 5: Heatmap of $Q_{T_p}(x_n)$ for $p = 3, \dots, 12$. The dashed line represents the critical relation $n = p \log_2(3)$.

This relation has a natural geometric interpretation. The finest dyadic scale present in x_n is of order 2^{-n} , up to a fixed shift arising from the indexing convention, whereas the mesh size of T_p is 3^{-p} . The two scales become comparable when

$$2^{-n} \asymp 3^{-p},$$

or equivalently,

$$n - p \log_2(3) = O(1).$$

We therefore introduce the rescaled variable

$$\xi = n - p \log_2(3).$$

Since

$$\frac{2^{-n}}{3^{-p}} = 2^{-\xi},$$

the variable ξ measures the relative separation between the dyadic scale of the truncation and the ternary scale of the partition.

After this rescaling, the increasing parts of the curves exhibit a clear approximate collapse onto a common transition profile. The location and shape of the transition become nearly independent of p .

When ξ is strongly negative, the finest dyadic scale present in x_n remains coarser than the ternary mesh, and the corresponding quadratic sum is small. As ξ approaches zero, the two scales become comparable and the quadratic sum increases rapidly. For positive ξ , finer dyadic levels are added and the curves approach their fixed- p limiting values.

The coarser cases $p \leq 3$ are omitted from the collapse plot because finite-resolution effects make their profiles less representative of the common transition region. The collapse concerns primarily the transition and does not imply that the limiting values are independent of p . A residual arithmetic dependence on the ternary refinement level remains visible after the transition.

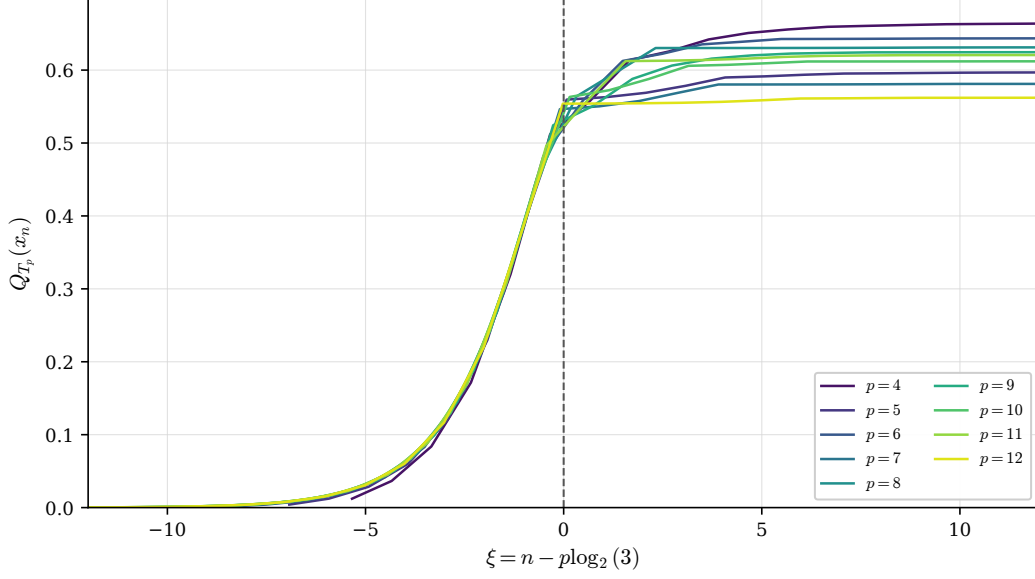


Figure 6: The quadratic sums $Q_{T_p}(x_n)$, for $p = 4, \dots, 12$, plotted against the rescaled variable $\xi = n - p \log_2(3)$. The dashed vertical line corresponds to $\xi = 0$.

The numerical evidence therefore indicates that the transition in $Q_{T_p}(x_n)$ is governed, to leading order, by

$$n - p \log_2(3),$$

rather than by n and p separately.

6.3 The full limiting function

We now return to the full constant-coefficient Takagi function x . For $p \leq 8$, the values of $Q_{T_p}(x)$ are obtained by directly evaluating the exact finite representation established in Section 5.

For $9 \leq p \leq 12$, direct evaluation of the complete finite formula becomes considerably more expensive because it contains

$$3^p H_p = 3^{2p-1}.$$

terms. We therefore use the partial sum x_{80} , together with an explicit bound on the resulting truncation error.

Indeed,

$$\|x - x_N\|_\infty \leq R_N, \quad R_N = \frac{2^{-N/2-1}}{1 - 2^{-1/2}}.$$

Consequently, every partition increment changes by at most $2R_N$. Writing

$$\delta_{p,j}^{(N)} = \Delta_{p,j}^T x_N,$$

we obtain

$$|Q_{T_p}(x) - Q_{T_p}(x_N)| \leq 4R_N \sum_{j=0}^{3^p-1} |\delta_{p,j}^{(N)}| + 4 \cdot 3^p R_N^2.$$

For $N = 80$ and $9 \leq p \leq 12$, this bound is smaller than

$$2.8 \times 10^{-9}.$$

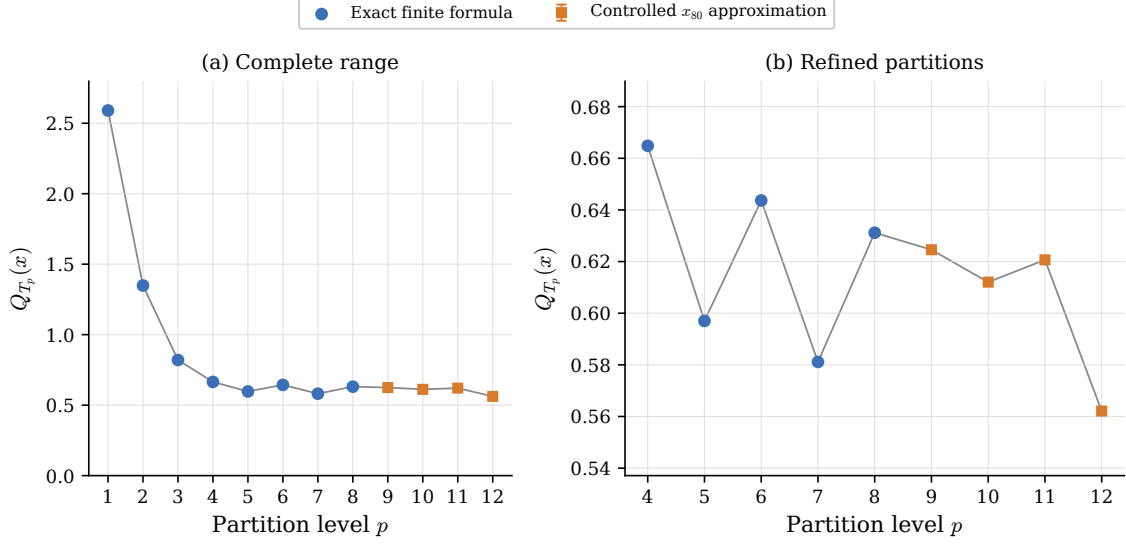


Figure 7: Computed values of $Q_{T_p}(x)$. Blue circles correspond to evaluations of the exact finite formula for $p \leq 8$. Orange squares correspond to the controlled x_{80} approximation for $9 \leq p \leq 12$. The left panel shows the complete range, while the right panel enlarges the region $p \geq 4$. The truncation-error bars are smaller than the displayed markers.

The first three values are comparatively large:

$$Q_{T_1}(x) \approx 2.5904120554, \quad Q_{T_2}(x) \approx 1.3488251111, \quad Q_{T_3}(x) \approx 0.8200694953.$$

These values reflect the coarse resolution of the first ternary partitions. From $p = 4$ onwards, the computed quadratic sums lie in the narrower interval

$$0.5621 \lesssim Q_{T_p}(x) \lesssim 0.6649.$$

The refined values remain non-negligible but display visible fluctuations. In particular, the computed sequence is not monotone, and the available data do not identify a convincing candidate for a limiting value.

It is important to distinguish two different limiting procedures. For each fixed p ,

$$Q_{T_p}(x_n) \longrightarrow Q_{T_p}(x) \quad \text{as } n \rightarrow \infty.$$

This does not imply that the sequence

$$(Q_{T_p}(x))_{p \geq 1}$$

converges as $p \rightarrow \infty$. The computations performed here are insufficient to determine whether the observed fluctuations eventually decay or persist. These questions are discussed in Section 7.

7 Discussion and Open Questions

This report has investigated how the quadratic sums of the normalised constant-coefficient Takagi function depend on the geometry of the underlying partitions. The comparison between dyadic and ternary partitions reveals two markedly different structures.

Along dyadic partitions, the Faber–Schauder expansion is exactly aligned with the partition grid. This compatibility produces orthogonality between the increment patterns associated with distinct dyadic levels and yields

$$Q_{D_p}(x) = 1 - 2^{-p}.$$

Consequently,

$$Q_D(x) = \lim_{p \rightarrow \infty} Q_{D_p}(x) = 1.$$

Along ternary partitions, the alignment is lost. Interior ternary partition points are not dyadic rationals, so the limiting function does not coincide with any finite partial sum at these points. Moreover, increments from distinct Faber–Schauder levels are no longer orthogonal when sampled on the ternary grid.

Despite these obstructions, the values of the dyadic orbit at a fixed ternary point possess an exact arithmetic structure. The multiplicative order

$$L_p = \text{ord}_{3^p}(2) = 2 \cdot 3^{p-1}$$

implies, through the evenness of the distance function, the shorter common period

$$H_p = 3^{p-1}.$$

This gives the exact finite representation

$$Q_{T_p}(x) = \frac{1}{(1 - 2^{-H_p/2})^2} \sum_{j=0}^{3^p-1} \left(\sum_{r=0}^{H_p-1} 2^{-r/2} d_{p,j}(r) \right)^2.$$

Thus, for every fixed p , the quadratic sum of the full limiting function can be evaluated without replacing x by an arbitrary partial sum. The original infinite-series problem is reduced to a finite arithmetic calculation involving

$$3^p H_p = 3^{2p-1}$$

terms.

The numerical investigation reveals a further scale-dependent phenomenon. For fixed p , the quadratic sums $Q_{T_p}(x_n)$ undergo a transition as n increases. The location of this transition is governed by the balance

$$2^{-n} \asymp 3^{-p},$$

or equivalently,

$$n - p \log_2(3) = O(1).$$

The approximate collapse of the transition curves under the rescaling

$$\xi = n - p \log_2(3)$$

provides numerical evidence that the relative position of the dyadic and ternary scales is the main parameter controlling the transition.

For the full limiting function, the computed values $Q_{T_p}(x)$ remain non-negligible over the range considered, but display visible arithmetic fluctuations. The numerical results establish neither convergence nor non-convergence of

$$(Q_{T_p}(x))_{p \geq 1}.$$

In particular, the convergence

$$Q_{T_p}(x_n) \longrightarrow Q_{T_p}(x) \quad \text{as } n \rightarrow \infty$$

for each fixed p must not be confused with the behaviour of $Q_{T_p}(x)$ as $p \rightarrow \infty$. These are two distinct limiting procedures.

7.1 Open questions

The principal question left unresolved by the present analysis is the asymptotic behaviour of

$$Q_{T_p}(x)$$

as $p \rightarrow \infty$. The available computations do not determine whether the sequence converges, possesses several subsequential limits, or continues to display persistent arithmetic fluctuations.

The finite representation obtained in Section 5 provides a possible starting point for this problem. Its direct evaluation requires 3^{2p-1} terms and therefore becomes increasingly expensive as p grows. A natural next step would be to search for additional cancellation or for a recursive relation connecting the representations at levels p and $p+1$.

The identities

$$H_{p+1} = 3H_p \quad \text{and} \quad 3^{p+1} = 3 \cdot 3^p$$

suggest that such a relation may be obtained by studying how residue classes modulo 3^{p+1} project onto residue classes modulo 3^p . A successful recurrence could provide both a more efficient numerical method and a route towards a rigorous asymptotic analysis.

A second direction would be to replace the ternary partitions by regular partitions in another odd base b . Since 2 is invertible modulo b^p , the corresponding dyadic orbit is again periodic. The multiplicative order

$$\text{ord}_{b^p}(2)$$

would then determine a common period, possibly shortened by symmetries of the underlying distance function. Comparing different bases could clarify which phenomena are specific to the pair $(2, 3)$ and which reflect a more general incompatibility between multiscale structures and partition grids.

It would also be natural to consider other choices of Faber–Schauder coefficients. The present report treats the constant-coefficient case

$$\theta_{m,k} = 1.$$

Allowing structured sign patterns or varying coefficients could reveal which aspects of the critical scaling and partition dependence persist for a broader class of deterministic multiscale functions.

More generally, this example illustrates that the quadratic variation of an irregular deterministic function need not be determined by the path alone. It may depend essentially on the interaction between the path and the sequence of partitions along which its increments are observed. The normalised constant-coefficient Takagi function provides an explicit setting in which this dependence can be analysed through a combination of real analysis, arithmetic periodicity and numerical computation.

Acknowledgements

I am grateful to Professor Rama Cont for his supervision, guidance and insightful discussions throughout this research internship.

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